**ID: 3cdbf026**

The graph of the equation $ax + ky = 6$ is a line in the *xy*-plane, where *a* and *k* are constants. If the line contains the points $(-2, -6)$ and $(0, -3)$, what is the value of *k* ?

A. $-2$

B. $-1$

C. 2

D. 3

ID: 9bbce683

| $x$ | $y$ |
|---|---|
| $18$ | $130$ |
| $23$ | $160$ |
| $26$ | $178$ |

For line $h$, the table shows three values of $x$ and their corresponding values of $y$. Line $k$ is the result of translating line $h$ down $5$ units in the *xy*-plane. What is the *x*-intercept of line $k$?

A. $\left(-\frac{26}{3}, 0\right)$

B. $\left(-\frac{9}{2}, 0\right)$

C. $\left(-\frac{11}{3}, 0\right)$

D. $\left(-\frac{17}{6}, 0\right)$

ID: db422e7f

Line $p$ is defined by $4y + 8x = 6$. Line $r$ is perpendicular to line $p$ in the *xy*-plane. What is the slope of line $r$?

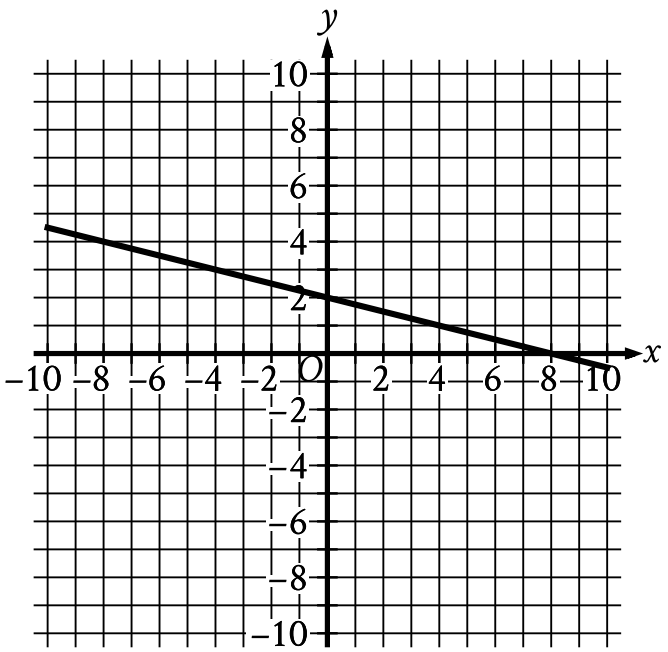
**ID: fdee0fbf**

In the *xy*-plane, line *k* intersects the *y*-axis at the point $(0, -6)$ and passes through the point $(2, 2)$. If the point $(20, w)$ lies on line *k*, what is the value of *w* ?

ID: b9835972

In the *xy*-plane, line $\ell$ passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 8x + 2$. If line $\ell$ also passes through the point $(3, d)$, what is the value of $d$?

ID: 05bb1af9




The graph of $y = f(x) + 14$ is shown. Which equation defines function $f$?

A. $f(x) = -\frac{1}{4}x - 12$

B. $f(x) = -\frac{1}{4}x + 16$

C. $f(x) = -\frac{1}{4}x + 2$

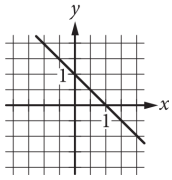
D. $f(x) = -\frac{1}{4}x - 14$

ID: 0b46bad5

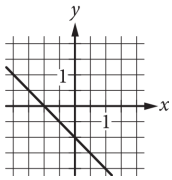
$$ax + by = b$$

In the equation above, *a* and *b* are constants and $0 < a < b$. Which of the following could represent the graph of the equation in the *xy*-plane?

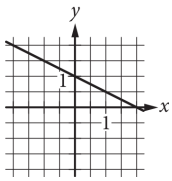
A.



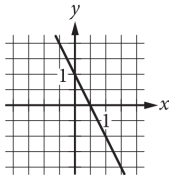

B.



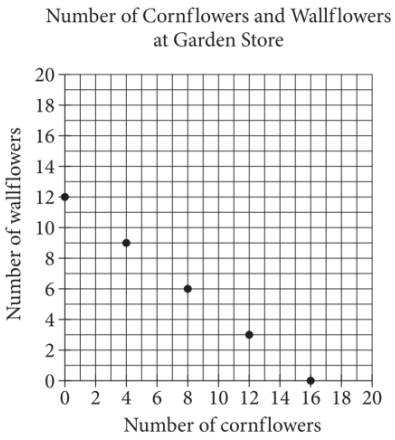

C.




D.



**ID: c362c210**




The points plotted in the coordinate plane above represent the possible numbers of wallflowers and cornflowers that someone can buy at the Garden Store in order to spend exactly $24.00 total on the two types of flowers. The price of each wallflower is the same and the price of each cornflower is the same. What is the price, in dollars, of 1 cornflower?

ID: 94b48cbf

The graph of $7x + 2y = -31$ in the $xy$-plane has an $x$-intercept at $(a, 0)$ and a $y$-intercept at $(0, b)$, where $a$ and $b$ are constants. What is the value of $\frac{b}{a}$?

A. $-\frac{7}{2}$

B. $-\frac{2}{7}$

C. $\frac{2}{7}$

D. $\frac{7}{2}$

ID: 98d3393a

Line $\ell$ in the *xy*-plane is perpendicular to the line with equation $x = 2$. What is the slope of line $\ell$?

A. 0

B. $-\frac{1}{2}$

C. $-2$

D. The slope of line $\ell$ is undefined.